[Untitled0.ipynb - Colab]

Q1. The operations on the GPU take place asynchronously. If we use torch.matmul(), then the operation will not wait for the GPU process to complete; instead, it will continue executing the next instructions. This will make the timer stop before its actual completion time, and hence an incorrect value for the execution time will be obtained. Synchronizing with torch.cuda.synchronize() makes the CPU wait until the GPU processes complete.

Q2. However, for extremely tiny matrices, the amount of computation done by the algorithm will not be enough to harness the parallel computing capabilities of the GPU. In fact, the time taken to call GPU kernel and transfer data to the GPU is much more than the computation. Hence, the CPU outperforms the GPU in such cases.